Transformata Fourier

Notăm $L^1(\mathbb{R}) = \{f: \mathbb{R} \to \mathbb{R} \mid \int_{-\infty}^{\infty} |f(t)|\,dt < \infty\}$

Def Fie $f \in L^1(\mathbb{R})$. Se numește transformata Fourier a funcției f, funcția $\mathcal{F}[f]: \mathbb{R} \to \mathbb{R}$ definită prin $\mathcal{F}[f](\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t}\,dt$

Notăm $\mathcal{F}[f](\omega) = F(\omega)$

$f(t) \xrightarrow{\mathcal{F}} F(\omega)$

Funcția $\mathcal{F}[f]$ se mai numește funcție spectrală sau spectrul în frecvență al semnalului $f(t)$

Dacă f e funcție pară, atunci

$$\mathcal{F}[f](\omega) = \int_{-\infty}^{\infty} f(t)\cdot e^{-i\omega t}\,dt = \int_{-\infty}^{\infty} f(t)(\cos\omega t - i\sin\omega t)\,dt$$

$$\mathcal{F}[f](\omega) = 2\int_{0}^{\infty} f(t)\cos\omega t\,dt$$ se numește transformata Fourier prin cosinus a lui f

Dacă f e funcție impară, atunci

$$\mathcal{F}[f](\omega) = -2i\int_0^\infty f(t)\sin\omega t\,dt$$

se numește transformata Fourier prin sinus a lui f

Formula Fourier de inversare

Fie $f \in L^1(\mathbb{R})$; $f(t) \xrightarrow{\mathcal{F}} F(\omega)$

Pp. $\int_{-\infty}^{\infty} |F(\omega)|\,d\omega < \infty$

atunci $f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)e^{i\omega t}\,d\omega$; $\forall t \in \mathbb{R}$

Proprietățile transformatei Fourier

Prop 1 → liniaritatea

Fie $f(t) \xrightarrow{\mathcal{F}} F(\omega)$ $\quad \alpha, \beta \in \mathbb{C}$

$g(t) \xrightarrow{\mathcal{F}} G(\omega)$

atunci $\alpha f(t) + \beta f(t) \xrightarrow{\mathcal{F}} \alpha F(\omega) + \beta G(\omega)$

Prop 2 → schimbare de scală

Fie $f(t) \xrightarrow{\mathcal{F}} F(\omega)$ și $a \in \mathbb{R}$

atunci $f(at) \xrightarrow{\mathcal{F}} \frac{1}{|a|}F\left(\frac{\omega}{a}\right)$

Prop 3 → **translația timpului**

Fie $f(t) \xrightarrow{\mathcal{F}} F(\omega)$ și $t_0 \in \mathbb{R}$

atunci $f(t-t_0) \xrightarrow{\mathcal{F}} e^{-it_0\omega} F(\omega)$

Prop 4 → **translația frecvenței**

Fie $f(t) \xrightarrow{\mathcal{F}} F(\omega)$ și $\omega_0 \in \mathbb{R}$

atunci $e^{i\omega_0 t} f(t) \xrightarrow{\mathcal{F}} F(\omega-\omega_0)$

Prop 5 → **derivarea în raport cu timpul**

Fie $f(t) \xrightarrow{\mathcal{F}} F(\omega)$

f derivabilă de n ori

atunci $f^{(n)}(t) \xrightarrow{\mathcal{F}} (i\omega)^n F(\omega)$

Prop 6 → **derivarea în raport cu frecvența**

Fie $f(t) \xrightarrow{\mathcal{F}} F(\omega)$

și funcțiile $tf(t)$; $t^2 f(t)$; ... $t^n f(t)$ sunt integrabile pe $\mathbb{R}$

atunci $(-it)^n f(t) \xrightarrow{\mathcal{F}} F^{(n)}(\omega)$

Prop 7 → convolutia în timp

Fie $f(t) \xrightarrow{\mathcal{F}} F(\omega)$

$g(t) \xrightarrow{\mathcal{F}} G(\omega)$

$(f*g)(t) = \int_{-\infty}^{\infty} f(\tau) g(t-\tau) d\tau$ produsul de convolutie al semnalelor f, g

atunci $(f*g)(t) \xrightarrow{\mathcal{F}} F(\omega)G(\omega)$

ex 1 Să se determine transformata Fourier pentru următorul semnal

$$f(t) = \left(1 - \frac{|t|}{a}\right)\left(u(t+2a) - u(t-2a)\right) \quad a>0$$

$$f(t) = \begin{cases} 1 - \frac{|t|}{a} & -2a < t < 2a \\ 0 & \text{în rest} \end{cases}$$

f pară $\Rightarrow \mathcal{F}[f](\omega) = 2\int_0^{\infty} f(t)\cos\omega t\, dt$

$$= 2\int_0^{2a}\left(1-\frac{t}{a}\right)\cos\omega t\, dt \underset{\omega\neq 0}{=\!=} 2\left[\left(1-\frac{t}{a}\right)\frac{\sin\omega t}{\omega}\Big|_0^{2a} + \frac{1}{a\omega}\int_0^{2a}\sin\omega t\, dt\right] = 2\left[-\frac{1}{\omega}\sin 2a\omega + \frac{1}{a\omega}\cdot\frac{-\cos\omega t}{\omega}\Big|_0^{2a}\right]$$

$$= 2\left[-\frac{1}{\omega}\sin 2a\omega + \frac{1}{a\omega^2}(\cos 2a\omega - 1)\right] \quad \omega \neq 0$$

$\cos 2a\omega - 1 = -2\sin^2 \omega a$

$$\mathcal{F}[f](0) = 2\int_0^{\infty} f(t)\cdot\cos 0\, dt = 2\int_0^{\infty} f(t)dt = 2\int_0^{2a}\left(1-\frac{t}{a}\right)dt$$

$$= 2\left(t-\frac{t^2}{2a}\right)\Big|_0^{2a} = 2\left(2a-0-\frac{4a^2}{2a}+0\right) = 2(2a-2a) = 0$$

$$\mathcal{F}[f](\omega) = \begin{cases} 2\left[-\frac{1}{\omega}\sin 2a\omega + \frac{1}{a\omega^2}\cdot 2\sin^2\omega a\right] & ;\ \omega \neq 0 \\ 0 & ;\ \omega = 0 \end{cases}$$

ex Să se determine transformata Fourier pentru următoarele semnale

a) $f(t) = t^2 e^{-3t}u(t) \xrightarrow[\text{deriv. în rap. cu frecv.}]{\mathcal{F}} -\mathcal{F}^{''}[e^{-3t}u(t)](\omega)$

$$\mathcal{F}[e^{-3t}u(t)](\omega) = \mathcal{F}[e^{-3t}u(-3t)](\omega) = \frac{1}{3}\cdot\mathcal{F}[e^{-t}u(t)]\left(\frac{\omega}{3}\right)$$

$$\mathcal{F}[e^{-t}u(t)](\omega) = \int_0^{\infty} e^{-t}e^{-i\omega t}dt = \int_0^{\infty} e^{-t(1+i\omega)}dt$$

$$= \frac{e^{-t(1+i\omega)}}{-(1+i\omega)}\Big|_0^{\infty} = \lim_{t\to\infty}\frac{e^{-t(1+i\omega)}}{-(1+i\omega)} + \frac{1}{1+i\omega} = 0 + \frac{1}{1+i\omega}$$

$e^{-t(1+i\omega)} \to$ constant pt. că $|1| = 1$

$$\mathcal{F}[e^{-3t}u(t)](\omega) = \frac{1}{3}\cdot\frac{1}{1+i\frac{\omega}{3}} = \frac{1}{3+i\omega}$$

$$-\mathcal{F}^{''}[e^{-3t}u(t)](\omega) = -\left(\frac{1}{3+i\omega}\right)^{''} = -\left(\frac{-i}{(3+i\omega)^2}\right)'$$

$$= \left(\frac{i}{(3+i\omega)^2}\right)' = \frac{-i\cdot 2(3+i\omega)\cdot i}{(3+i\omega)^4} = \frac{2}{(3+i\omega)^3}$$

b) $f(t) = e^{-4|2t+1|} = e^{-8|t+1/2|} \xrightarrow[\text{transl. timp.}]{\mathcal{F}}$

$\to e^{i\frac{\omega}{2}} \mathcal{F}[e^{-8|t|}](\omega) \xrightarrow[\text{de. scală}]{\text{schim.}} e^{i\frac{\omega}{2}} \cdot \frac{1}{8} \cdot \mathcal{F}[e^{-1|t|}]\left(\frac{\omega}{8}\right)$

$\mathcal{F}[e^{-1|t|}](\omega) = \int_{-\infty}^{\infty} \underbrace{e^{-1|t|}}_{\text{pară}} \cdot e^{-i\omega t}\, dt = 2\int_0^{\infty} e^{-t}\cos\omega t\, dt$

$\mathcal{L}[\cos\omega t](1)$

$= 2 \cdot \mathcal{L}[\cos\omega t](1) = 2 \cdot \frac{1}{1+\omega^2} = \frac{2}{1+\omega^2}$

$\mathcal{F}[f(t)](\omega) = \frac{1}{8} \cdot e^{i\frac{\omega}{2}} \cdot \frac{2}{1+\frac{\omega^2}{64}} = \frac{16}{64+\omega^2} \cdot e^{i\frac{\omega}{2}}$

c) $f(t) = e^{-|t-1|}\sin(t-1) \xrightarrow[\text{transl. timp.}]{\mathcal{F}} e^{-i\omega} \cdot \mathcal{F}[e^{-1|t|}\sin t](\omega)$

$\mathcal{F}[g(t)\sin\alpha t](\omega) = \mathcal{F}\left[g(t) \cdot \frac{e^{i\alpha t} - e^{-i\alpha t}}{2i}\right](\omega) \xrightarrow{\text{liu}}$

$= \frac{1}{2i}\left[\mathcal{F}[e^{i\alpha t}g(t)](\omega) - \mathcal{F}[e^{-i\alpha t}g(t)](\omega)\right] \xrightarrow[\text{frecv.}]{\text{transl.}}$

$= \frac{1}{2i}\left[\mathcal{F}[g(t)](\omega-\alpha) - \mathcal{F}[g(t)](\omega+\alpha)\right]$

$\mathcal{F}[g(t)\cos\alpha t](\omega) = \mathcal{F}\left[g(t) \cdot \frac{e^{i\alpha t} + e^{-i\alpha t}}{2}\right](\omega) \xrightarrow{\text{liu}}$

$= \frac{1}{2}\left[\mathcal{F}[e^{i\alpha t}g(t)](\omega) + \mathcal{F}[e^{-i\alpha t}g(t)](\omega)\right] \xrightarrow[\text{frecv.}]{\text{transl.}}$

$= \frac{1}{2}\left[\mathcal{F}[g(t)](\omega-\alpha) + \mathcal{F}[g(t)](\omega+\alpha)\right]$

$$\mathcal{F}[f(t)](\omega) = e^{-i\omega} \cdot \frac{1}{2i}\left[\mathcal{F}[e^{-|t|}](\omega-1) - \mathcal{F}[e^{-|t|}](\omega+1)\right]$$

$$\underline{\text{ex 1 b}}\quad \frac{1}{2i} e^{-i\omega}\left[\frac{2}{1+(\omega-1)^2} - \frac{2}{1+(\omega+1)^2}\right]$$

d) $f(t) = e^{-3(t-5)^2} \xrightarrow[\text{transl. timp.}]{\mathcal{F}} e^{-5i\omega}\mathcal{F}[e^{-3t^2}](\omega)$

$$\underset{\text{de scal.}}{\overset{\text{schim}}{}} e^{-5i\omega} \cdot \frac{1}{\sqrt{3}}\mathcal{F}[e^{-t^2}]\left(\frac{\omega}{\sqrt{3}}\right) = \frac{1}{\sqrt{3}} e^{-5i\omega}\cdot\sqrt{4}\cdot e^{-\frac{\omega^2}{12}}$$

② Calculați $f*g$ și $\mathcal{F}[f*g](\omega)$ pentru

$f(t) = g(t) = e^{-t^2}$, $t \in \mathbb{R}$ semnalul gaussian

$$\rightarrow (f*g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t-\tau)\,d\tau = \int_{-\infty}^{\infty} e^{-\tau^2}\cdot e^{-(t-\tau)^2}\,d\tau$$

$$= \int_{-\infty}^{\infty} e^{-\tau^2-t^2+2t\tau-\tau^2}\,d\tau = e^{-t^2}\int_{-\infty}^{\infty} e^{-2\tau^2+2t\tau}\,d\tau$$

$$= e^{-t^2}\int_{-\infty}^{\infty} e^{-2(\tau^2-t\tau)}\,d\tau = e^{-t^2}\int_{-\infty}^{\infty} e^{-2(\tau^2-2\cdot\frac{1}{2}t\tau+\frac{1}{4}t^2-\frac{1}{4}t^2)}\,d\tau$$

$$(f*g)(t) = e^{-t^2/2}\int_{-\infty}^{\infty} e^{-2(\tau-\frac{t}{2})^2}\,d\tau$$

Notăm $\sqrt{2}(\tau-\frac{t}{2}) = \theta \Rightarrow d\theta = \sqrt{2}\,d\tau$

$$(f*g)(t)=e^{-t^2/2}\int_{-\infty}^{\infty}e^{-\theta^2}\frac{d\theta}{\sqrt{2}}=\frac{e^{-t^2/2}}{\sqrt{2}}\int_{-\infty}^{\infty}e^{-\theta^2}d\theta$$

$$(f*g)(t)=\frac{e^{-t/2}}{\sqrt{2}}\sqrt{\pi}$$

$\longrightarrow$ $\mathcal{F}[f*g](\omega)=\mathcal{F}[f](\omega)\cdot\mathcal{F}[g](\omega)$

$$F(\omega)=\mathcal{F}[f](\omega)=\mathcal{F}[g](\omega)=\mathcal{F}[e^{-t^2}](\omega)=$$

$$=\int_{-\infty}^{\infty}e^{-t^2}e^{-i\omega t}dt=2\int_0^{\infty}e^{-t^2}\cos\omega t\,dt$$

para

$$\Rightarrow F'(\omega)=2\int_0^{\infty}e^{-t^2}(-t\sin\omega t)dt=\int_0^{\infty}-2te^{-t^2}\sin\omega t\,dt$$

$$=e^{-t^2}\sin\omega t\Big|_0^{\infty}-\omega\int_0^{\infty}e^{-t^2}\cos\omega t\,dt=-\omega\frac{F(\omega)}{2}$$

$t\to 0$ (0) — $\sin\omega t$ vale 0

$$F'(\omega)=\frac{-\omega}{2}F(\omega)\Rightarrow\frac{F'(\omega)}{F(\omega)}=\frac{-\omega}{2}\Rightarrow \ln F=\frac{-\omega^2}{4}+C$$

$$\Rightarrow F(\omega)=e^{-\frac{\omega^2}{4}+C}=K\cdot e^{-\frac{\omega^2}{4}}$$

$$F(0)=2\int_0^{\infty}e^{-t^2}dt=2\cdot\frac{\sqrt{\pi}}{2}=\sqrt{\pi}\not\Rightarrow K=\sqrt{\pi}$$

$$F(\omega)=\sqrt{\pi}\cdot e^{-\frac{\omega^2}{4}}$$

$F(0)=K$

$$\mathcal{F}[f*g](\omega)=\sqrt{\pi}\cdot\left(e^{-\frac{\omega^2}{4}}\right)^2=\sqrt{\pi}\cdot e^{-\frac{\omega^2}{2}}$$

③ Cu ajutorul transformatei Fourier să se rezolve următoarea ecuație integrală

a) $\int_0^\infty f(\alpha)\sin(\alpha x)\,d\alpha = \underbrace{e^{-x}}_{F}$; $x>0$

Prelungim funcția f prin imparitate pe $\mathbb{R}$.

Analog F $\Rightarrow$

$$\Rightarrow \int_{-\infty}^{\infty} f(\alpha)\cos(\alpha x)\,d\alpha = 0$$

$$\int_{-\infty}^{\infty} f(\alpha)\sin(\alpha x)\,d\alpha = 2\int_0^\infty f(\alpha)\sin(\alpha x)\,d\alpha = 2F(x)\big|(-i)$$

$$\Rightarrow \int_{-\infty}^{\infty} f(\alpha)\underbrace{[\cos(\alpha x) - i\sin(\alpha x)]}_{e^{-i\alpha x}}\,d\alpha = -2iF(x)$$

$$\Rightarrow \underbrace{\int_{-\infty}^{\infty} f(\alpha)e^{-i\alpha x}\,d\alpha}_{\mathcal{F}[f(\alpha)](x)} = -2iF(x) \xrightarrow[\text{Fourier de invers.}]{\text{form.}}$$

$$\longrightarrow f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} -2iF(x)\underbrace{e^{i\alpha x}}_{\cos\alpha x + i\sin\alpha x}dx = \frac{-i}{\pi}2i\int_0^\infty F(x)\sin(\alpha x)dx$$

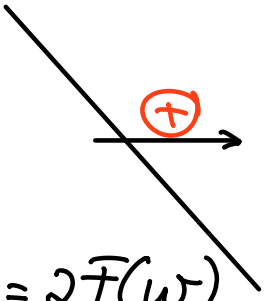
$$= \frac{2}{\pi}\int_0^\infty e^{-x}\sin(\alpha x)\,dx = \frac{2}{\pi}\cdot\frac{\alpha}{1+\alpha^2}$$

$$\mathcal{L}[\sin(\alpha x)](1) = \frac{\alpha}{1+\alpha^2}$$

b) $\int_0^\infty f(t)\cos\omega t\,dt = \frac{1}{(1+\omega^2)^2} \overset{\text{Not.}}{=\!=} F(\omega)$

Prelungim funcția f prin paritate pe $\mathbb{R}$. $\Rightarrow$

$\Rightarrow \int_{-\infty}^{\infty} f(t)\sin(\omega t)\,dt = 0 \quad |\cdot(-i)$



$\int_{-\infty}^{\infty} f(t)\cos\omega t\,dt = 2\int_0^{\infty} f(t)\cos\omega t\,dt = 2F(\omega)$

$\int_{-\infty}^{\infty} f(t)e^{-i\omega t} = 2F(\omega) \xrightarrow[\text{de inv.}]{\text{form. Fourier}}$

$\mathcal{F}[f(t)](\omega)$

$\Rightarrow f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} 2F(\omega)e^{i\omega t}\,d\omega$

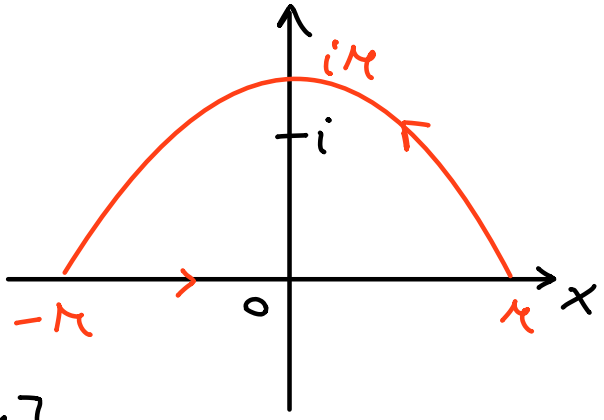
$\cos\omega t + i\sin\omega t \searrow 0$

$\Rightarrow f(t) = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{1}{(1+\omega^2)^2}\cos\omega t\,d\omega$

$\rightarrow$ calculăm $\int_{-\infty}^{\infty} \frac{1}{(1+\omega^2)^2}e^{i\omega t}\,dt$

Fie $g(z) = \frac{e^{izt}}{(1+z^2)^2}$

$\{\pm i\}$ poli de ordinul 2



Alegem $R > 1 \quad \gamma = \gamma_1 \cup [-R, R]$

$\int_{\gamma_2} g(z)dz + \int_{-R}^{R} g(z)dz = \int_{\gamma} g(z)dz \overset{\text{th. rez.}}{=} 2\pi i \operatorname{Rez}(g(z), i)$

$$\operatorname{Rez}(g(z), i) = \lim_{z \to i} \left((z-i)^2 \frac{e^{izt}}{(z-i)^2 (z+i)^2} \right)'$$

$$= \lim_{z \to i} \frac{it e^{izt}(z+i)^2 - e^{izt} 2(z+i)}{(z+i)^4} = \frac{it e^{-t} \cdot 2i - e^{-t} \cdot 2}{(2i)^3}$$

$$= \frac{-2te^{-t} - 2e^{-t}}{-8i} = \frac{te^{-t} + e^{-t}}{4i}$$

$$\int_\gamma g(z)dz = 2\pi i \cdot \frac{te^{-t} + e^{-t}}{4i} = \frac{\pi}{2} e^{-t}(t+1)$$

$$\int_\gamma g(z)dz = \int_{-\infty}^{\infty} \frac{1}{(1+\omega^2)^2} e^{i\omega t} dt = \frac{\pi}{2} e^{-t}(t+1)$$

$$\Rightarrow f(t) = \frac{1}{\pi} \cdot \frac{\pi}{2} e^{-t}(t+1) = \frac{1}{2} e^{-t}(t+1)$$